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S2.3 The metallic model

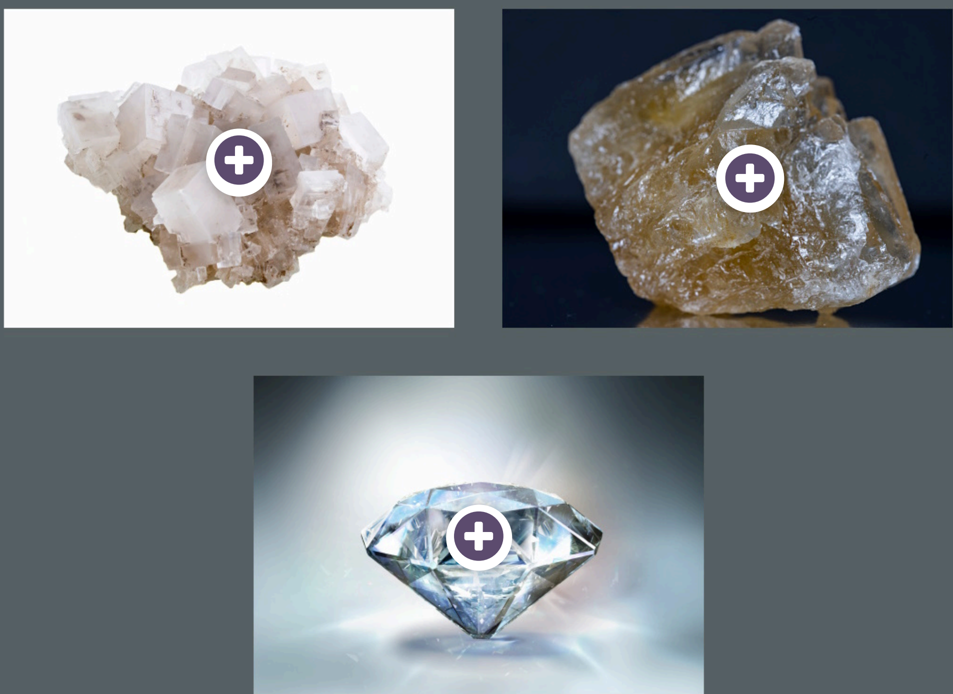
Metallic bonding and structure

Learning outcomes

By the end of this section you should be able to:

- Define metallic bonding.
- Describe the structure of the metallic lattice.
- Sketch a diagram showing the lattice structure of a metal.

When we think of crystalline solids, we typically think of ionic and covalent solids (e.g. sodium chloride, sugar and diamond). A crystalline solid is where the constituents are arranged in a highly ordered lattice structure that extends in all directions. **Interactive 1** shows some of these examples of ionic and covalent crystals.



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Interactive 1. Crystalline Solids of Sodium Chloride, Sugar and Diamond.

More information for interactive 1

An interactive collage displays three photographs of crystalline solids. Each photograph has a plus sign representing an interactive hotspot. Selecting a hotspot reveals the internal crystal structure of the corresponding material.
Top left:
Photo: A chunk of rock salt displaying intergrown white to transparent crystals.
Hotspot: A cubic structure of sodium chloride featuring an alternating arrangement of sodium and chloride ions. Chloride ions are represented by large spheres, and sodium ions are represented by small spheres.
Top right:
Photo: An irregularly shaped golden brown sugar crystal.
Hotspot: The molecular structure of sugar depicts repeating sucrose units with carbon, hydrogen, and oxygen atoms.
Bottom:
Photo: A single crystal of diamond displaying its shiny, reflective surface.
Hotspot: A cubic structure illustrates carbon atoms occupying the corners and face centers of the cube. Each carbon atom is bonded to four other carbon atoms in a tetrahedral geometry.
This interactive collage helps users identify and recognize the external appearance and internal structure of three crystalline substances.

We have seen examples of crystalline substances that have ionic and covalent bonding. Did you know that metals also exist as crystalline solids? **Figure 1** shows an example of bismuth metal that crystallises in a unique staircase structure. As you learn more about metals, consider the structure and bonding of metals compared to ionic salts and covalent network crystals.

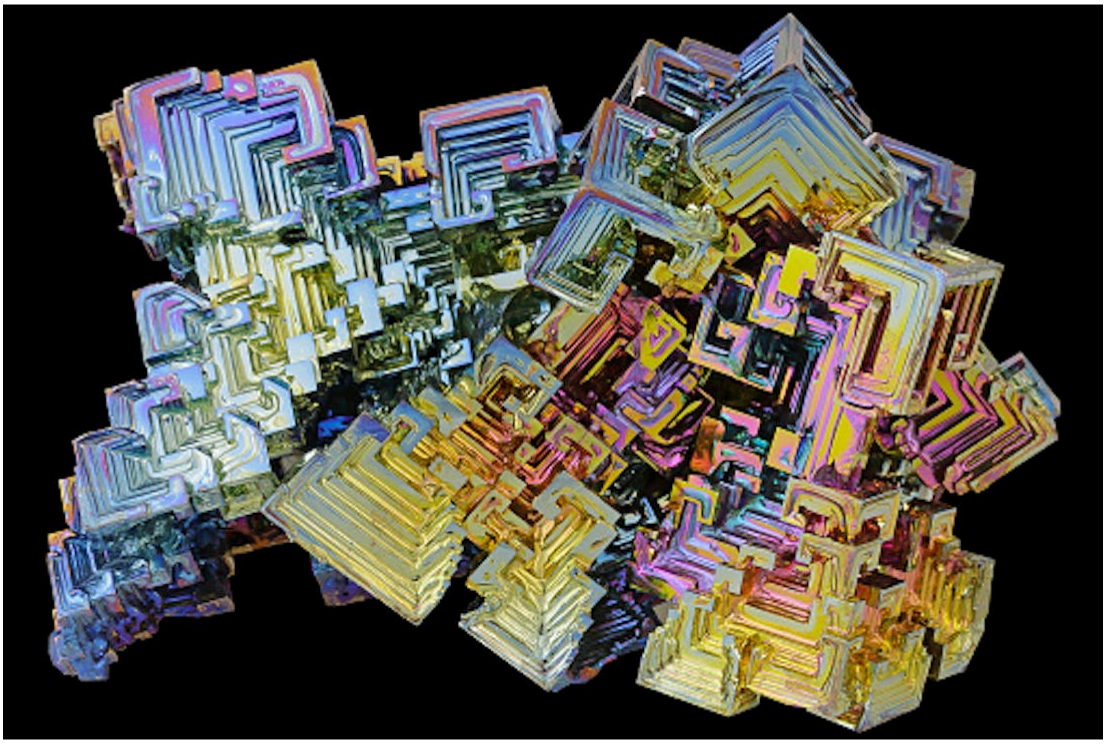


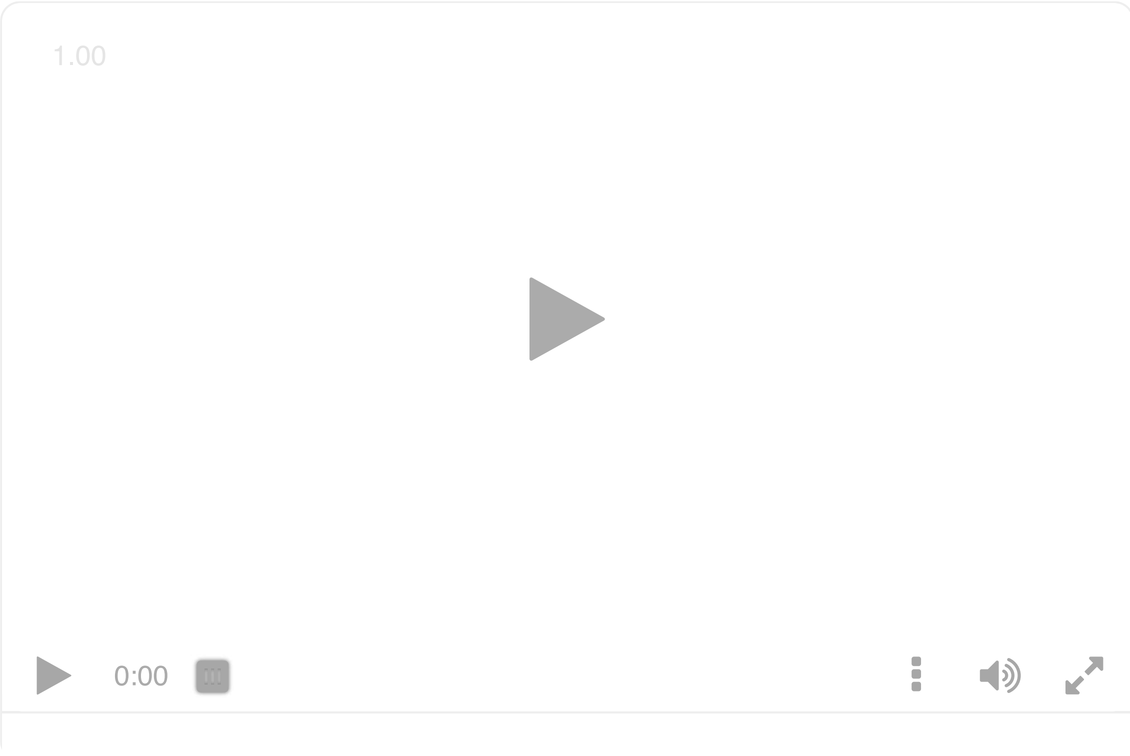
Figure 1. Bismuth crystals.

Credit: Walter Geiersperger, Getty Images

(https://www.gettyimages.co.uk/license/589145472)

In section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) you learned how ionic bonding resulted from the electrostatic attraction between oppositely charged ions. Metals are neutral atoms. How are uncharged atoms able to attract one another to form a bond? What could be occurring within the metallic bond to allow neutral particles to remain attracted together as a solid?

Let us look at the formation of the metallic bond by considering the metallic element lithium in **Interactive 2**.



Interactive 2. Formation of the Metallic Bond for Lithium.

More information for interactive 2

A video displays four rows and four columns of lithium atoms. The text on the left reads as follows: Solid lithium exists in a metallic lattice. But how do these atoms stay together? Let's look closer at lithium's electrons.

In the next clip, a single Li atom is depicted. On-screen text reads: Electron configuration: $1s^2\,2s^1$. $1s^2$ are the core electrons. $2s^1$ is the valence electron. When metal atoms are arranged in a lattice, the outer valence electrons are not fixed to an individual lithium atom. Instead, these valence electrons are shared with all the atoms in the metallic lattice.

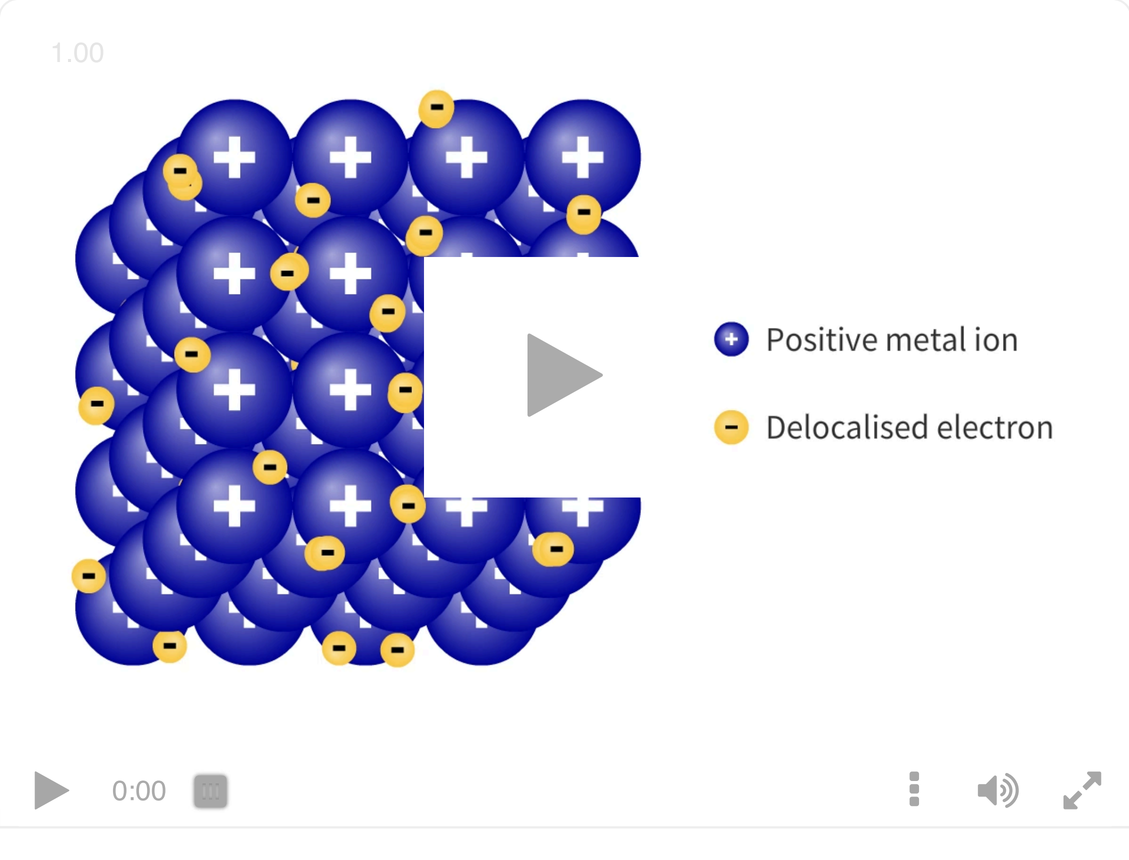
Next clip: A lithium ion with a positive charge and a valence electron outside it. The valence electron is indicated using an arrow. The text reads as follows: This makes the metal atoms in the lattice positively charged. The valence electron is free to move throughout the lattice, meaning it is delocalised.

Next clip: A lattice with four rows and four columns of Li^+ ions, each with a valence electron around it. The Li^+ ions are represented as circles, and the electrons are represented as dots. The text reads as follows: The metallic lattice is made up of metal cations arranged in a regular pattern with delocalised valence electrons freely moving between them.

Next clip: Four Li^+ ions with their valence electrons. Double-headed arrows are placed between the ions and nearby valence electrons. The text on the left reads as follows: There is a strong electrostatic attraction between positive metal ions and delocalised valence electrons. This electrostatic attraction is how metal atoms stay together in a lattice. This is the metallic bond.

The interactive helps users understand the formation of metallic bond in lithium atoms.

When metal atoms arrange in a lattice, the outer valence electrons are strongly attracted to neighbouring atoms, moving through the empty spaces in the lattice away from the atom it was originally part of. These electrons are shared among all metal atoms in the lattice. As the valence electrons are not bound to any one specific atom they are called delocalised electrons and free to move throughout the lattice. As valence electrons are delocalised, the remaining metal atoms in the lattice become positively charged cations. These cations repel each other, because they have the same charge, and keep the neatly arranged lattice in place. There are very strong electrostatic forces between the lattice of positive metal cations and the delocalised electrons: this is the metallic bond and is shown in **Interactive 3**.



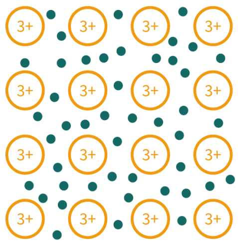
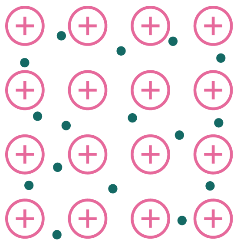
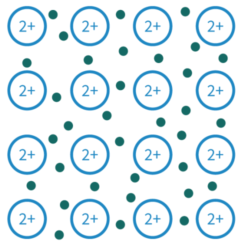
Interactive 3. Structure of Metals.

More information for interactive 3

An interactive video exhibits a cubic metal lattice structure with positive metal ions arranged in an orderly manner. Positive ions are indicated as purple circles with a plus sign in the middle. Delocalized electrons move randomly through the empty spaces within the lattice. These electrons are indicated as yellow circles with a minus sign in the middle. Delocalized electrons, also known as valence electrons, are shared among all the metal ions in the lattice.
The "symbols" to identify the positive and negative ions in the lattice are provided on the right side of the lattice structure.
The play button with a slider allows the users to play and pause the video as required. In the bottom right, the speedometer, sound, setting, and zoom-out icons are provided consecutively. Except for the setting icon, all other icons are highlighted in white.
This video helps users understand the structure of the metal and the movement of delocalized electrons.

In **Interactive 4**, drag and drop the box with the electron configuration, number of valence electrons and the structure into the correct location in the table.

Metal	Sodium	Aluminium	Magnesium
Electron configuration			
# valence electrons			
Structure			

$1s^22s^22p^63s^2$	$1s^22s^22p^63s^1$	$1s^22s^22p^63s^23p^1$
1	2	3
		

Check

Interactive 4. Electron Configurations and Valence Electrons for Different Structures.

More information for interactive 4

An interactive table features labels on the top and the left with missing fields. The labels on the top from left to right read: Metal, Sodium, Aluminum, and Magnesium. The labels on the left from top to bottom read: Electron configuration, # valence electrons, and Structure. There are three missing rows and columns in the interactive table.
Below the table, there are three rows and three columns with draggable options. The options in the first row from left to right include: $1s^22s^22p^63s^2$, $1s^22s^22p^63s^1$, and

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$1s^2 2s^2 2p^6 3s^2 3p^1$. The options in the second row from left to right include: 1, 2, and 3. The options in the third row from left to right include:
A square box with orange circles (metal ions) labeled with 3+ that are surrounded by a sea of electrons in peacock green.
A square box with pink circles (metal ions) labeled with + that are surrounded by a sea of electrons in peacock green.
A square box with blue circles (metal ions) labeled with 2+ that are surrounded by a sea of electrons in peacock green.
This interactivity helps users understand metallic structures, valence electrons, and electron configurations for sodium, aluminum, and magnesium. The “Check” icon at the bottom left allows users to check their answers. The right answer is indicated with a slider. If the inputted answers are wrong, the “Retry” icon at the bottom left allows users to retry the activity.
Read below for the solution.

Sodium

Electronic configuration: $1s^2 2s^2 2p^6 3s^1$
Number of valence electrons: 1
Structure: A square box with pink circles (metal ions) labeled with + that are surrounded by a sea of electrons in peacock green.

Aluminum

Electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^1$
Number of valence electrons: 3
Structure: A square box with orange circles (metal ions) labeled with 3+ that are surrounded by a sea of electrons in peacock green

Magnesium

Electronic configuration: $1s^2 2s^2 2p^6 3s^2$
Number of valence electrons: 2
Structure: A square box with blue circles (metal ions) labeled with 2+ that are surrounded by a sea of electrons in peacock green.

Exercise 1

Click a question to answer

1. True or false?

Metallic bonding is the electrostatic attraction between cations in the three-dimensional lattice.

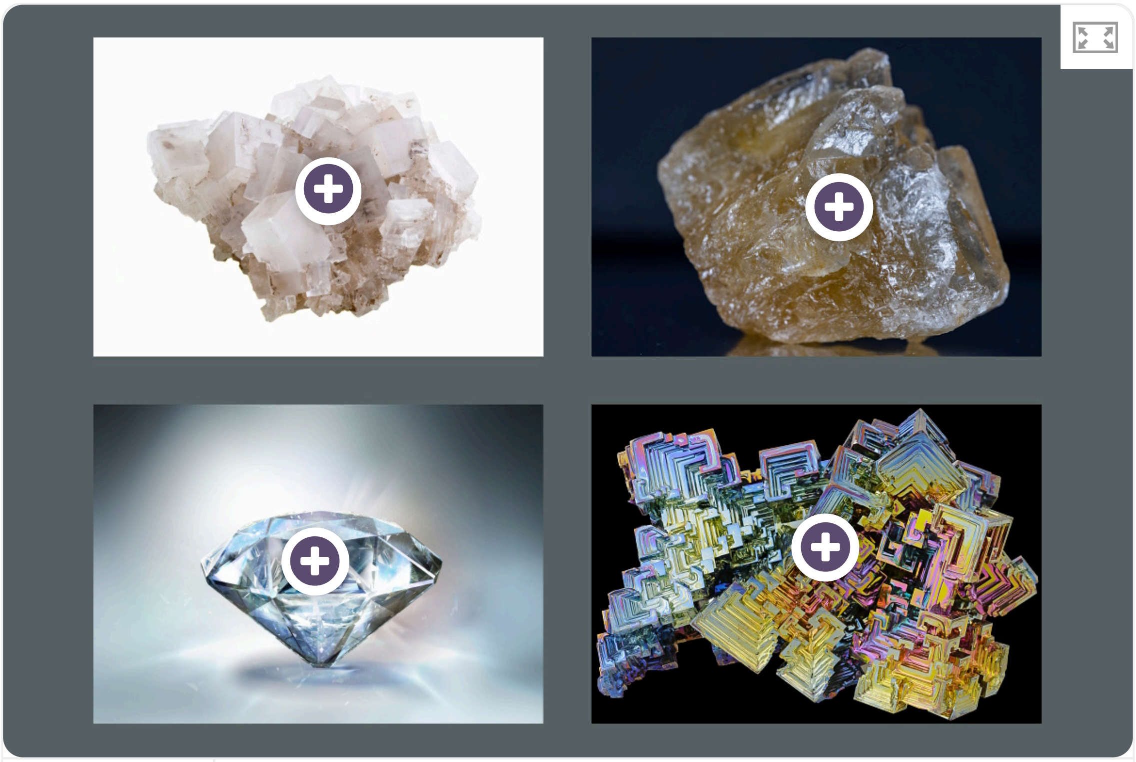
Check answers

Nature of Science

Aspect: Models

This metallic structure and bonding model is known as the 'free electron model'. While it is incredibly useful to help understand the physical properties of metals through considering ions at the atomic scale, this model also has its limitations. For example, the repulsive interaction between the valence electrons is mostly neglected. One way to identify unbiased information is the acknowledgement of limitations.

Metallic bonding is unique from its ionic and covalent counterparts as the bonding is not between atoms/ions. Instead, the bonding occurs between the cations within the lattice structure and the delocalised valence electrons. **Interactive 5** shows a comparison of the structures for all three types of bonding. These valence electrons act similarly to a glue keeping the cations bonded together. This type of bonding is non-directional because the electrostatic attractive force occurs in all directions.



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Interactive 5. Crystalline Solids of Sodium Chloride, Sugar, Diamond and Bismuth.

More information for interactive 5

An interactive collage displays four photographs of crystalline solids. Each photograph has a plus sign representing an interactive hotspot. Selecting a hotspot reveals the type of bonding and the internal structure of the corresponding material.
Top left:
Photo: A chunk of rock salt displaying intergrown white to transparent crystals.
Hotspot: Ionic. A cubic structure of sodium chloride featuring an alternating arrangement of sodium and chloride ions. Chloride ions are represented by large spheres, and sodium ions are represented by small spheres.
Top right:
Photo: An irregularly shaped golden brown sugar crystal.
Hotspot: Covalent-small molecule. The molecular structure of sugar depicts repeating sucrose units with carbon, hydrogen, and oxygen atoms.
Bottom left:
Photo: A single crystal of diamond displaying its shiny, reflective surface.
Hotspot: Covalent network. A cubic structure illustrates carbon atoms occupying the corners and face centers of the cube. Each carbon atom is bonded to four other carbon atoms in a tetrahedral geometry.
Bottom right:
Photo: A bismuth crystal exhibits step-like structures and iridescent colors. The crystal is shiny and has sharp edges.
Hotspot: Metallic: A cubic structure of arranged positive metal ions surrounded by a sea of delocalized negative electrons.
This interactive collage helps users identify the types of bonding in crystalline solids and their internal arrangements.

It is important to note that although the structure of metals involves metal cations, metals themselves are electrically neutral because all valence electrons remain part of the overall structure.

Exercise 2

Click a question to answer

1. Which of the following species are involved in metallic bonding:

I. Cations
II. Core electrons
III. Valence electrons

Check answers

Activity

- **IB learner profile attribute**: Communicator
- **Approaches to learning:**
  - Thinking skills — Applying key ideas and facts in new contexts
  - Communication skills — Clearly communicating complex ideas in response to open-ended questions
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair activity

Balloon bonding models. Each student will need one balloon.

In pairs, use your balloons to model the three types of bonding. Your balloon will represent a valence electron and your body represents the nucleus of an atom. The combination of you and your balloon is neutral.

- How might you use the balloons to demonstrate ionic bonding with your partner? What process needs to occur with neutral atoms before an ionic compound can be formed?
- How might you use the balloons to demonstrate covalent bonding with your partner? What occurs between two atoms when a covalent bond is formed?
- How might you use the balloons to demonstrate metallic bonding? What are valence electrons doing in a metal?

Discuss the following with your partner:

- In each type of bonding, identify and explain the source of attraction between particles. How did you show this with your balloon model?

Compare and contrast metallic bonding with the other types of bonding using a Venn diagram:

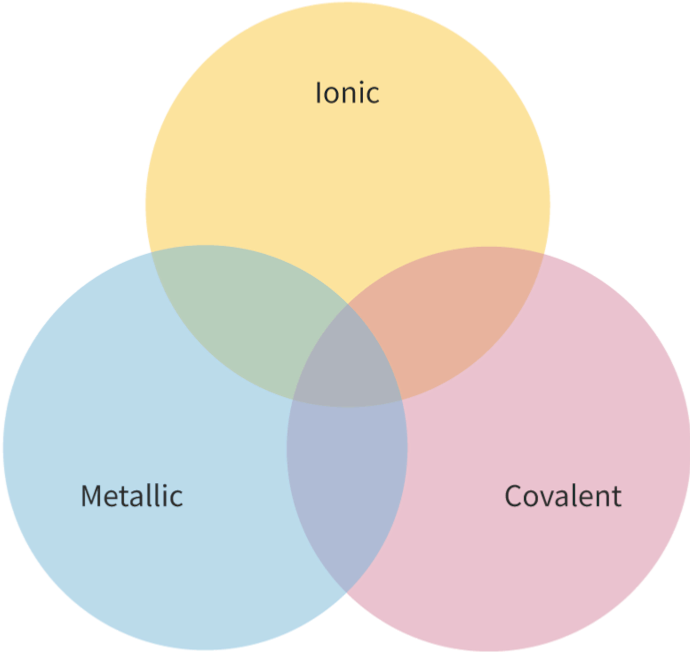

Figure 2. Venn diagram for comparing and contrasting metallic bonding with the other types of bonding.

More information for figure 2

The image is a Venn diagram that compares and contrasts three types of chemical bonding: metallic, ionic, and covalent. The diagram consists of three intersecting circles. Each circle represents one type of bonding. The first circle, labeled 'Metallic,' overlaps with the other two circles. The second circle, labeled 'Ionic,' overlaps with the 'Metallic' and 'Covalent' circles. The third circle is labeled 'Covalent.' The overlaps in the Venn diagram represent shared characteristics between the bonding types. For example, where the 'Metallic' and 'Ionic' circles overlap, common properties shared by metallic and ionic bonding are indicated. The central area, where all three circles intersect, would show characteristics that all three types of bonding share. However, the specific shared characteristics are not listed in the diagram and require external knowledge or context to fill in.

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Extension

- **IB learner profile attribute:** Communicator
- **Approaches to learning:**
  - Thinking skills — Applying key ideas and facts in new contexts
  - Communication skills — Clearly communicating complex ideas in response to open-ended questions

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- **Time required to complete activity:** 1 hour
- **Activity type:** Group/pair activity

Create a stop motion film to show the structure of a metallic lattice with delocalised electrons. Your target audience are primary school students. Use any material you have available to do so. There are some easy-to-find websites (e.g. Flipgrid (https://my.flipgrid.com/)) and social media tools that you can use to upload images and create a stop motion film.

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S2.3 The metallic model

Strength of the metallic bond

Learning outcomes

By the end of this section you should be able to:

- Predict the relative strength of the metallic bond by considering the charge and radius of the metal ion.
- Explain trends in melting points of s and p block metals.

Recall that a metallic bond is the electrostatic attraction between cations in a lattice and delocalised electrons, as illustrated in **Figure 1** below. By considering the particles that make up the metallic bond, can you hypothesise what factors would influence its strength? What factors might lead to a strong metallic bond? What factors might lead to a weak metallic bond? How would you expect the strength of a metallic bond to impact the melting point of a metal?

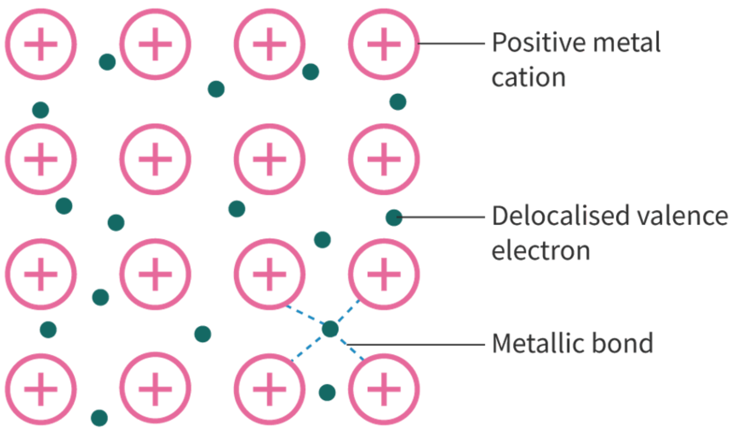


Figure 1. Metallic bond.

More information for figure 1

The diagram illustrates the structure of a metallic bond. It shows an orderly array of pink circles, each marked with a plus sign representing positive metal cations. Interspersed between these cations are small green circles representing delocalized valence electrons. Arrows and labels point out the metallic bond, which is the interaction between the cations and the electrons. The metallic bond is characterized by lines connecting the electrons to the surrounding cations, indicating the electrostatic attraction.

[Generated by AI]

Strength of the metallic bond

The strength of metallic bonds varies depending on the nature of the metal. The strength of the electrostatic force of attraction between delocalised electrons and cations within the metallic lattice is dependent on the same factors as those affecting ionic bonding, as in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/). Those factors are radius of the metal ion and ionic charge, which in combination determines the electron density. Let us consider the reasoning behind why these factors both influence metallic bond strength below.

Radius of metal ion

When you looked at the subatomic composition of the atom in section S1.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/) you may have noticed that the atom is mostly empty space. What do you think determines the radius of a metal cation? The protons in the nucleus could be very close or very far away from the valence shell electrons that delocalise through the metallic lattice. Do you think larger or smaller cations would have a stronger attractive force with the delocalised electrons in a metallic lattice?

The smaller the radius of the metal ion, the stronger the metallic bond. This is because of the shorter distance between the positive nucleus of the cation and the surrounding delocalised electrons.

Let us consider the strength of the metallic bond by considering the electron configuration and ionic radius for group 1 metals.

Table 1. Electron configuration and ionic radii for group 1 metals.

Element	Electron configuration	Charge of cation	Ionic radius (10^{-1}
Li	$[He]2s^1$	1+	76
Na	$[Ne]3s^1$	1+	102
K	$[Ar]4s^1$	1+	138
Rb	$[Kr]5s^1$	1+	152
Cs	$[Xe]6s^1$	1+	167

For group 1 metals, each atom loses one electron to the delocalised electrons in the lattice. Group 1 metal ions will have the same charge (1+) and the same number of delocalised electrons per metal atom. Metals further down the group will have a larger cationic radius due to the presence of an additional energy level.

As lithium metal ions have a smaller radius, the lithium nucleus will be closer to the delocalised electrons than the other group 1 metals and will result in a greater electrostatic attraction. This greater attraction means a stronger metallic bond, requiring more energy to break. Lithium, therefore, has the strongest metallic bond of the group 1 metals. This is illustrated in **Figure 2** below.

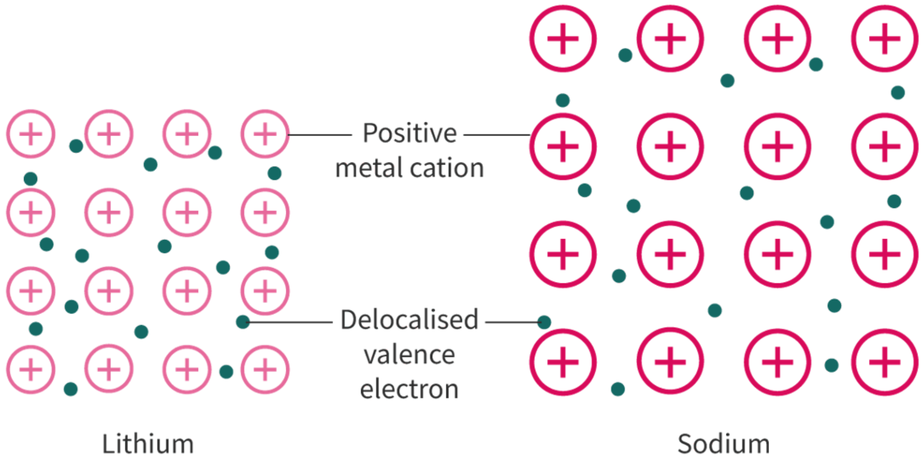


Figure 2. Comparison of the metallic bond of lithium and sodium.

More information for figure 2

The diagram compares the metallic bond structure of lithium and sodium. On the left, lithium is shown with a denser arrangement of positive metal cations and delocalized valence electrons. Positive metal cations are represented as circles with a positive sign inside, which are closely packed. Delocalized valence electrons are shown as small dots scattered between the cations. The labeling indicates that the distance between lithium's nucleus and its delocalized electrons is smaller, leading to a greater electrostatic attraction.

On the right, sodium is displayed with a similar structure but with a looser packing of positive metal cations and delocalized valence electrons, illustrating a lower electrostatic attraction compared to lithium. The comparison visually demonstrates why lithium forms a stronger metallic bond than sodium due to its smaller atomic radius, resulting in a greater attraction between the ions and electrons.

[Generated by AI]

Charge of the metal ion

Recall that the metallic bond is the electrostatic attraction between cations in a lattice and delocalised electrons. The charge of the cations is dependent on the number of electrons in the valence shell. How do you think the charge on the metal ion would impact the attractive forces with the delocalised electrons in the metallic lattice?

The higher the ionic charge, the stronger the metallic bond. This is because:

- greater charge on the metal ion

- greater number of delocalised valence electrons.

This greater charge difference results in a stronger electrostatic attraction and a stronger metallic bond.

Let us consider period 3 metals sodium and magnesium as an example.

Table 2. Electron configuration and charge for sodium and magnesium.

Element	Electron configuration	Charge of cation
Na	$[Ne]3s^1$	1+
Mg	$[Ne]3s^2$	2+

The ionic radius sizes for these two metal ions will be similar as they have the same number of occupied energy levels. Because magnesium has a 2+ charge, it will have a greater number of electrons making up the surrounding delocalised electrons than sodium. This greater charge difference increases the strength of the electrostatic attraction between metal ions and delocalised electrons. Magnesium, therefore, has a stronger metallic bond than sodium. This is illustrated in **Figure 3** below.

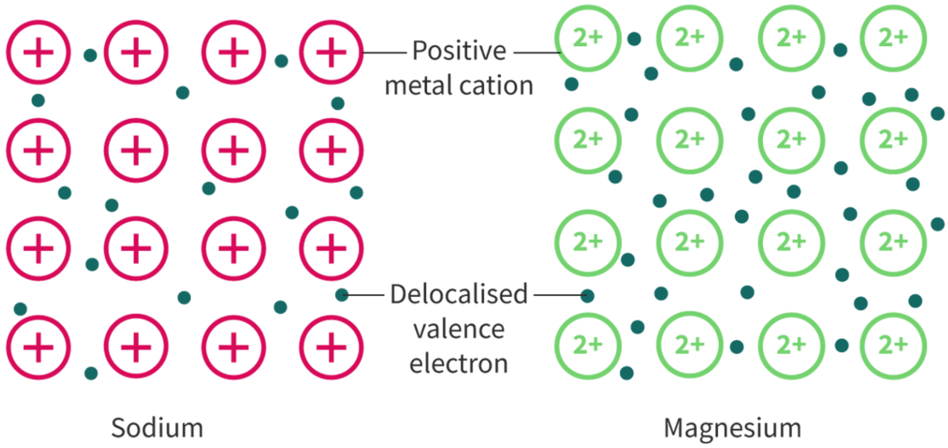


Figure 3. Comparison of the metallic bond of sodium and magnesium.

More information for figure 3

The image is a diagram comparing the metallic bonding structures of sodium and magnesium. On the left, sodium is depicted with '+' symbols inside circles representing positive metal cations, surrounded by small dots that represent delocalised valence electrons. On the right, the magnesium structure is similar but with '2+' inside the circles, indicating a stronger positive charge resulting in more delocalised electrons around the metal cations. This visual highlights the difference in the strength of the metallic bond, with magnesium showing a stronger attraction due to the increased charge and electron density.

[Generated by AI]

In summary, the metallic bond becomes stronger with increased electron density. Decreasing ionic radius and increasing charge on the metal ion increases the relative number of electrons in a region of space. The smaller the radius and the greater the charge of the metal ion, the stronger the metallic bond. The larger the radius and the lower the charge of the metal ion, the weaker the metallic bond. Try out **Interactive 1** below to see the effect of radius and charge of the metal ion on the strength of the metallic bond.

Adjust the radius or charge bar for Metal 2 and look at what happens to the metallic lattice image when you adjust the size, charge and the number of electrons.

Interactive 1. Effect of radius and charge of the metal ion on the strength of the metallic bond.

More information for interactive 1

An interactive simulation features two metals, Metal 1 and Metal 2. The cations in both metals initially have a +1 charge and the same number of delocalized electrons per metal atom within their metallic lattice.

Radius and Charge slider bars are at the bottom right of metal 2. The radius or the charge for Metal 2 can be manipulated to identify its influence on the metallic bond.

On moving the radius slider, the ionic radius of Metal 2 increases, and the displayed text reads Metal 1 has stronger metallic bonds than Metal 2. This explains how the increase in ionic radius affects the metallic bond.

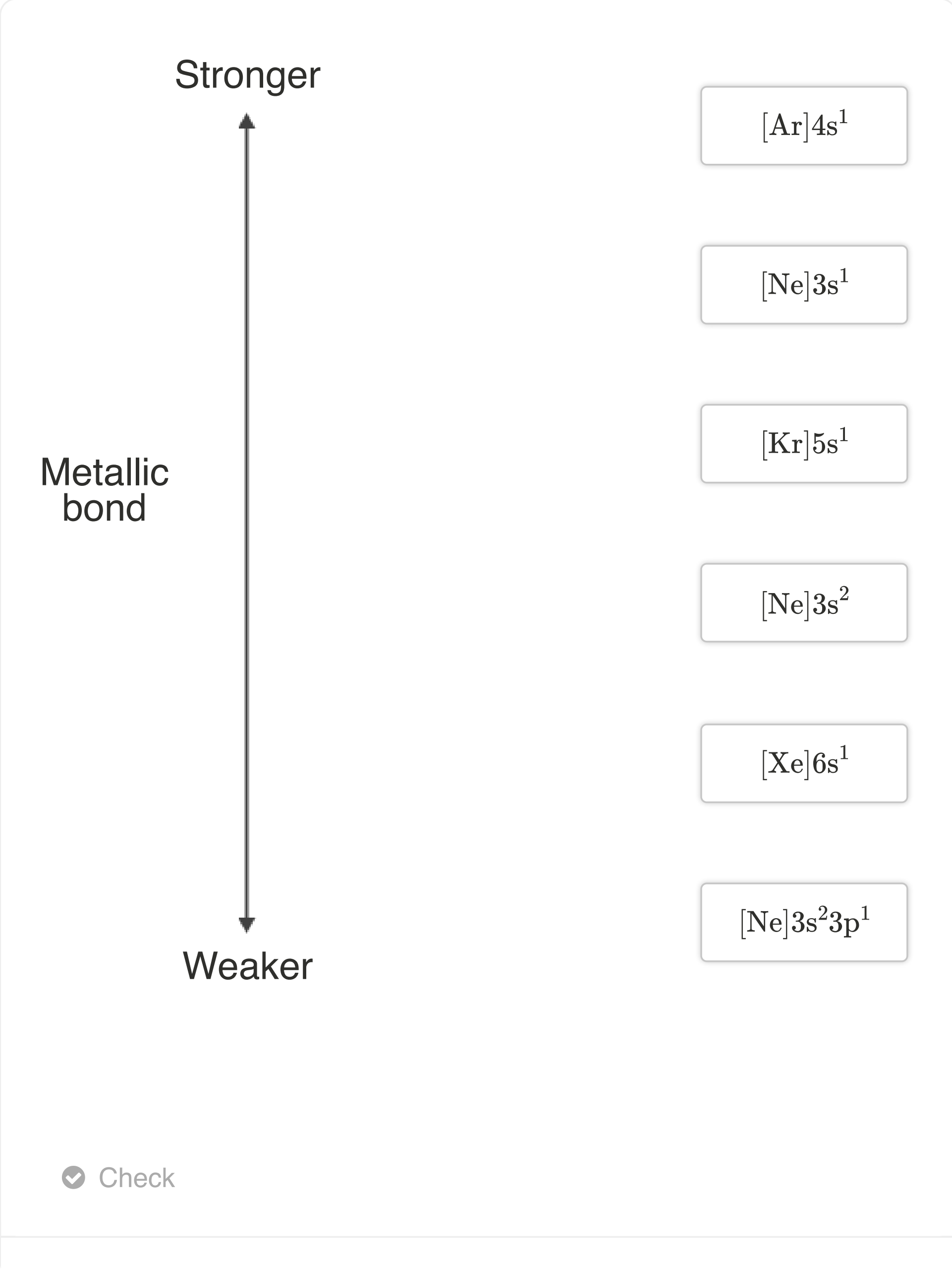
On moving the charge slider, the ionic charge of Metal 2 increases with the increase in the number of delocalized electrons, and the displayed text reads, Metal 1 has weaker metallic bonds than Metal 2. This explains how the increase in charge affects the metallic bond.

This interaction helps users understand the influence of metal ion radius and charge on metallic bond strength.

Making connections

The strength of the metallic bond is dependent on the same factors as the strength of the ionic bond, as explored in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/). The strength of both bonds depends on ionic radius and ionic charge.

Check your understanding of the strength of metallic bonds with **Interactive 2** by dragging and dropping the boxes of different electron configurations of metals and placing them in order of the strength of metallic bond from strongest to weakest.



Interactive 2. Understanding the Strength of Metallic Bonds.

Strength of metallic bond and melting points

The strength of metallic bonds results in high melting points for metals. From your experience, what state of matter are metals at room temperature? Do you know of any metal that exists in a different state at room temperature? Why might that be?

At room temperature, all metals, except mercury, are solids. For metals found in the s and p block regions of the periodic table, the melting point is directly related to the strength of the metallic bond. Note that the metals in the d block are not considered in this section due to the added complexity when considering the valence electrons in d orbitals. Melting points for all elements can be found in section 8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/) of the DP chemistry data booklet.

Figure 4 shows a periodic table with melting points for the s and p block metals highlighted. Looking at groups 1 and 2, what conclusion could you draw about the melting point of metals as you go down the group? Why do you think this is? Looking at period 3, what conclusion could you draw about the melting point of metals as you travel across a period? Why do you think this is?

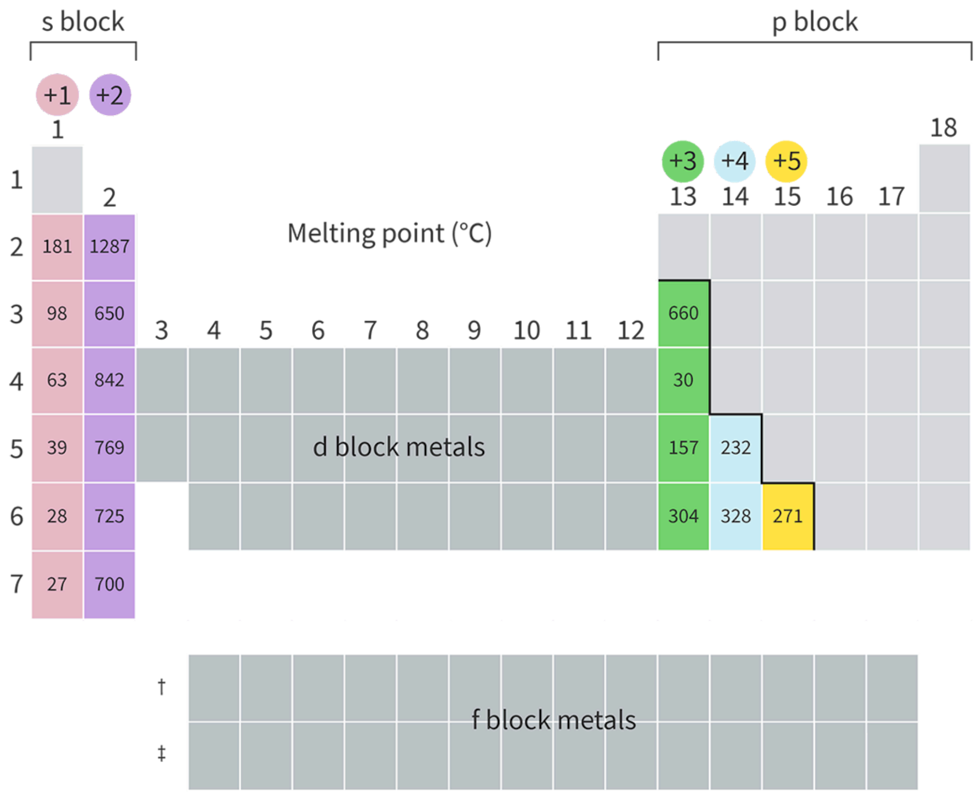


Figure 4. Melting points for the s and p block metals.

More information for figure 4

The image is a modified periodic table highlighting the melting points of s and p block metals. The table showcases groups 1 and 2, labeled with their respective changes in melting points as you move down the groups. For instance, Group 1 has melting points such as 181, 98, 63, 39, 28, and 27 moving downwards, indicating a general decrease. In Group 2, the melting points are 1287, 650, 842, 769, 725, and 700 moving downwards, showing variations. The s block metals are highlighted on the left side in shades of pink and purple. The p block metals on the right side include melting points such as 660, 30, 157, 304, 328, 232, and 271 in green, blue, and yellow blocks for typical groups +3, +4, and +5 respectively. The d and f blocks are shown in grey but not highlighted as they are not the primary focus. This visual representation indicates that melting points decrease downwards in groups and vary across periods due to differences in metallic bonding and electron density.

[Generated by AI]

As you can see, the melting point decreases as you go down a group of elemental metals. This is due to the decreased strength of the metallic bond as the radius of the metal ion increases. Generally, across a period, for elemental metals, the melting point increases. This is due to the increased strength of the metallic bond due to the increase in electron density as the charge on the metal ion becomes larger.

You might notice that the trend in melting point is much less distinct across a period than down a group. If you look at period 4, gallium is a significant outlier with a very low melting point: to the extent that objects made from gallium can melt when placed in the palm of your hand, or in warm water, as shown in **Video 1** below.

Video 1. Gallium spoon melting.

More information for video 1

The video presents a demonstration of a physical property of gallium metal. A transparent glass mug filled halfway with water is placed on a wooden table, alongside a polished, silver-coloured teaspoon made of gallium.
A hand enters the frame and lifts the spoon, initially holding it horizontally before rotating it to a vertical position. The hand then carefully lowers the spoon into the glass of water, immersing it. As the immersion progresses, the spoon is swirled around in the water. Over time, subtle changes become visible: small droplets begin to form along the spoon's surface. These droplets gradually increase in size before detaching and falling away, revealing that the spoon is undergoing a change. Rather than remaining solid, the metal appears to be melting. Eventually, the entire spoon melts completely, disappearing into the water.
The interactive video illustrates the low melting point of gallium as the metal transitions from a solid to a liquid when immersed in a cup with water. Gallium has a very low melting point, and when placed in water that is even slightly warm or on the palm of a hand, it begins to liquefy.
By showcasing the transformation of gallium in real time, the video highlights how physical properties such as melting point can vary significantly between different elements.

Theory of Knowledge

Mercury was consumed by many powerful historical figures, believing that it would grant immortality. Today, we have a greater understanding of this mysterious substance as a lethal neurotoxin, resulting in symptoms such as insomnia, emotional changes and disturbances in sensations. Does our current knowledge of this dangerous substance change our perspective of historical events?

Study skills

While there are trends for melting point in certain regions of the periodic table, there are not any trends for melting point across the periodic table. This is important to consider. The melting point trends discussed here are valid for s and p block metals, but do not apply to non-metals. It is important to consider the structure and bonding of elements to explain any trends.

Strength of metallic bond and hardness

The strength of metallic bonds results in a greater hardness of the metal. Hardness is a measure of the ability of a material to resist deformation. Think of the metals you have encountered at home: would you consider them hard or soft? What do you think that implies about the strength of their metallic bonds? Did you know that some metals are so soft they can be cut with a knife? **Interactive 3** shows two group 1 metals being sliced, sodium and potassium. Before watching them be cut, make a prediction. Which metal do you expect to be harder and have stronger metallic bonds? Which metal do you expect to be softer and have weaker metallic bonds? Then watch each metal being cut to see if you can identify the softer metal (weaker metallic bonds).

Interactive 3. Cutting Sodium and Potassium, a Measure of Hardness.

More information for interactive 3

The interactive contains two videos titled Sodium and Potassium. The video on the left for sodium is titled Sodium metal is soft and squishy. The video demonstrates the physical properties of sodium metal. The sodium metal chunks are stored in a glass container filled with oil. A chunk of sodium is removed from the oil and is pressed with fingers, revealing no signs of crumbling or breaking. The chunk is then cut using a knife. The video shows that sodium is soft enough to be easily cut with a knife, revealing its shiny interior. The cut piece is stored in the glass container with oil again. The video on the right for potassium is titled Potassium metal is like weird butter. The video demonstrates the physical properties of potassium metal. Potassium metal is taken out of a small tube and is then cut with a knife. The video shows that potassium is soft and can be easily cut like butter, revealing a shiny interior. The metal is pressed with fingers, revealing signs of crumbling and breaking. The cut piece of metal, when added to a container with water, reacts immediately emitting smoke and flame. The interactive demonstrates the metallic nature of sodium and potassium.

Activity

- **IB learner profile attribute**: Communicators
- **Approaches to learning**: Communication skills — Clearly communicating complex ideas in response to open-ended questions
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** Individual activity

Spreadsheet plotting

In this activity you will be creating individual plots to show how the radius of the metal ion affects the melting point for group 1, group 2 and group 13 metals.

- Predict what you expect the shape of each graph to look like for each group. Create a sketch of what you think each graph will look like.
- Create individual plots showing the effect of the radius of the metal ion on the melting point for group 1, group 2 and group 13 metals using information found in section 8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/) of the DP chemistry data booklet.
- For each plot, consider the following questions:
  - Did this plot look similar to your sketch? Why or why not?
  - Are there any data points that did not align with your expectations? Try researching to see whether you can come up with any explanations.
  - How do the melting points of s and p block metals provide evidence for the strength of the metallic bond?

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S2.3 The metallic model

Strength of the metallic bond

Learning outcomes

By the end of this section you should be able to:

- Predict the relative strength of the metallic bond by considering the charge and radius of the metal ion.
- Explain trends in melting points of s and p block metals.

Recall that a metallic bond is the electrostatic attraction between cations in a lattice and delocalised electrons, as illustrated in **Figure 1** below. By considering the particles that make up the metallic bond, can you hypothesise what factors would influence its strength? What factors might lead to a strong metallic bond? What factors might lead to a weak metallic bond? How would you expect the strength of a metallic bond to impact the melting point of a metal?

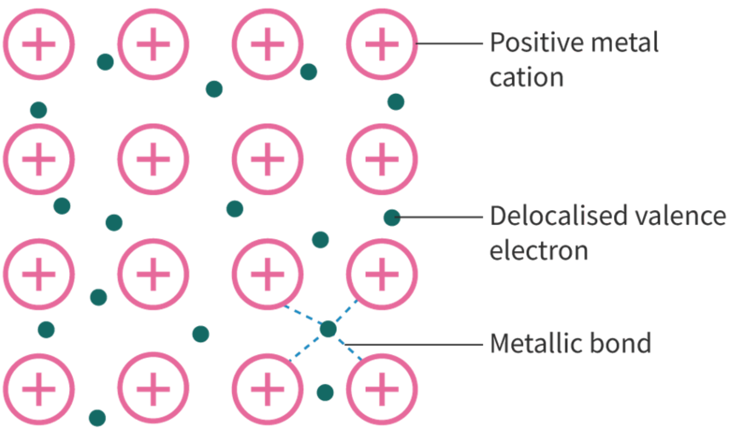


Figure 1. Metallic bond.

More information for figure 1

The diagram illustrates the structure of a metallic bond. It shows an orderly array of pink circles, each marked with a plus sign representing positive metal cations. Interspersed between these cations are small green circles representing delocalized valence electrons. Arrows and labels point out the metallic bond, which is the interaction between the cations and the electrons. The metallic bond is characterized by lines connecting the electrons to the surrounding cations, indicating the electrostatic attraction.

[Generated by AI]

Strength of the metallic bond

The strength of metallic bonds varies depending on the nature of the metal. The strength of the electrostatic force of attraction between delocalised electrons and cations within the metallic lattice is dependent on the same factors as those affecting ionic bonding, as in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/). Those factors are radius of the metal ion and ionic charge, which in combination determines the electron density. Let us consider the reasoning behind why these factors both influence metallic bond strength below.

Radius of metal ion

When you looked at the subatomic composition of the atom in section S1.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/) you may have noticed that the atom is mostly empty space. What do you think determines the radius of a metal cation? The protons in the nucleus could be very close or very far away from the valence shell electrons that delocalise through the metallic lattice. Do you think larger or smaller cations would have a stronger attractive force with the delocalised electrons in a metallic lattice?

The smaller the radius of the metal ion, the stronger the metallic bond. This is because of the shorter distance between the positive nucleus of the cation and the surrounding delocalised electrons.

Let us consider the strength of the metallic bond by considering the electron configuration and ionic radius for group 1 metals.

Table 1. Electron configuration and ionic radii for group 1 metals.

Element	Electron configuration	Charge of cation	Ionic radius (10^{-1}
Li	$[He]2s^1$	1+	76
Na	$[Ne]3s^1$	1+	102
K	$[Ar]4s^1$	1+	138
Rb	$[Kr]5s^1$	1+	152
Cs	$[Xe]6s^1$	1+	167

For group 1 metals, each atom loses one electron to the delocalised electrons in the lattice. Group 1 metal ions will have the same charge (1+) and the same number of delocalised electrons per metal atom. Metals further down the group will have a larger cationic radius due to the presence of an additional energy level.

As lithium metal ions have a smaller radius, the lithium nucleus will be closer to the delocalised electrons than the other group 1 metals and will result in a greater electrostatic attraction. This greater attraction means a stronger metallic bond, requiring more energy to break. Lithium, therefore, has the strongest metallic bond of the group 1 metals. This is illustrated in **Figure 2** below.

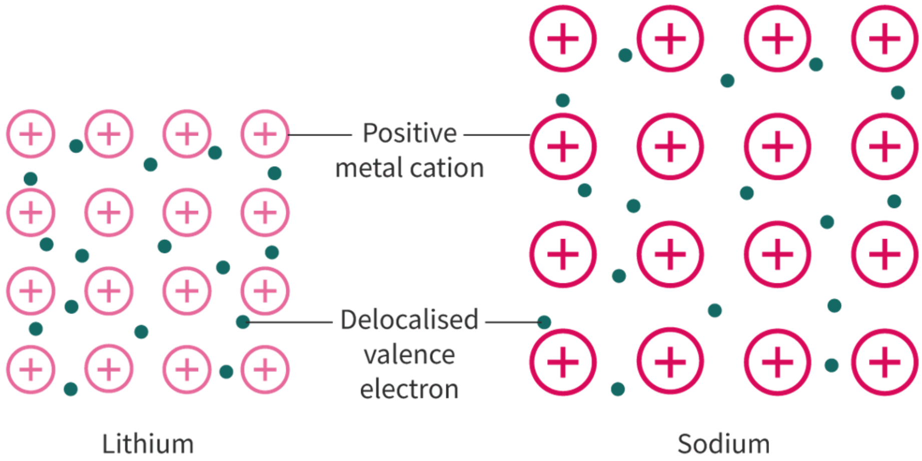


Figure 2. Comparison of the metallic bond of lithium and sodium.

More information for figure 2

The diagram compares the metallic bond structure of lithium and sodium. On the left, lithium is shown with a denser arrangement of positive metal cations and delocalized valence electrons. Positive metal cations are represented as circles with a positive sign inside, which are closely packed. Delocalized valence electrons are shown as small dots scattered between the cations. The labeling indicates that the distance between lithium's nucleus and its delocalized electrons is smaller, leading to a greater electrostatic attraction.

On the right, sodium is displayed with a similar structure but with a looser packing of positive metal cations and delocalized valence electrons, illustrating a lower electrostatic attraction compared to lithium. The comparison visually demonstrates why lithium forms a stronger metallic bond than sodium due to its smaller atomic radius, resulting in a greater attraction between the ions and electrons.

[Generated by AI]

Charge of the metal ion

Recall that the metallic bond is the electrostatic attraction between cations in a lattice and delocalised electrons. The charge of the cations is dependent on the number of electrons in the valence shell. How do you think the charge on the metal ion would impact the attractive forces with the delocalised electrons in the metallic lattice?

The higher the ionic charge, the stronger the metallic bond. This is because:

- greater charge on the metal ion

- greater number of delocalised valence electrons.

This greater charge difference results in a stronger electrostatic attraction and a stronger metallic bond.

Let us consider period 3 metals sodium and magnesium as an example.

Table 2. Electron configuration and charge for sodium and magnesium.

Element	Electron configuration	Charge of cation
Na	[Ne]$3s^1$	1+
Mg	[Ne]$3s^2$	2+

The ionic radius sizes for these two metal ions will be similar as they have the same number of occupied energy levels. Because magnesium has a 2+ charge, it will have a greater number of electrons making up the surrounding delocalised electrons than sodium. This greater charge difference increases the strength of the electrostatic attraction between metal ions and delocalised electrons. Magnesium, therefore, has a stronger metallic bond than sodium. This is illustrated in **Figure 3** below.

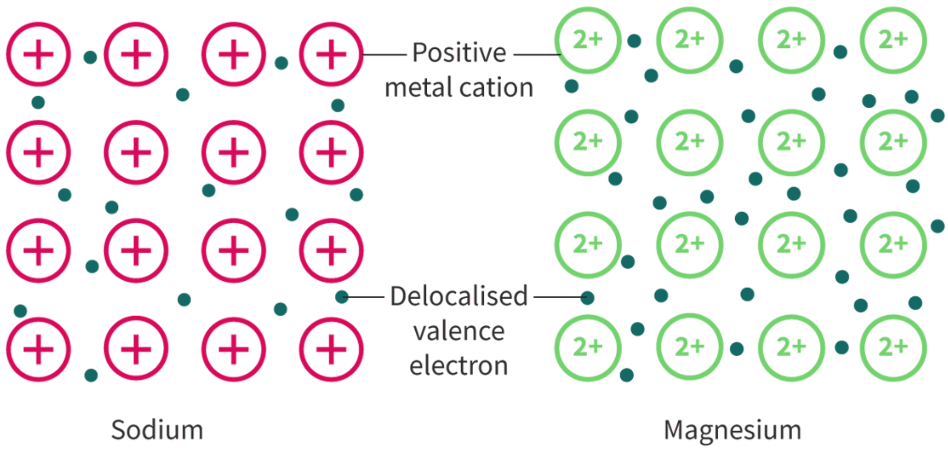


Figure 3. Comparison of the metallic bond of sodium and magnesium.

More information for figure 3

The image is a diagram comparing the metallic bonding structures of sodium and magnesium. On the left, sodium is depicted with '+' symbols inside circles representing positive metal cations, surrounded by small dots that represent delocalised valence electrons. On the right, the magnesium structure is similar but with '2+' inside the circles, indicating a stronger positive charge resulting in more delocalised electrons around the metal cations. This visual highlights the difference in the strength of the metallic bond, with magnesium showing a stronger attraction due to the increased charge and electron density.

[Generated by AI]

In summary, the metallic bond becomes stronger with increased electron density. Decreasing ionic radius and increasing charge on the metal ion increases the relative number of electrons in a region of space. The smaller the radius and the greater the charge of the metal ion, the stronger the metallic bond. The larger the radius and the lower the charge of the metal ion, the weaker the metallic bond. Try out **Interactive 1** below to see the effect of radius and charge of the metal ion on the strength of the metallic bond.

Adjust the radius or charge bar for Metal 2 and look at what happens to the metallic lattice image when you adjust the size, charge and the number of electrons.

Interactive 1. Effect of radius and charge of the metal ion on the strength of the metallic bond.

More information for interactive 1

An interactive simulation features two metals, Metal 1 and Metal 2. The cations in both metals initially have a +1 charge and the same number of delocalized electrons per metal atom within their metallic lattice.

Radius and Charge slider bars are at the bottom right of metal 2. The radius or the charge for Metal 2 can be manipulated to identify its influence on the metallic bond.

On moving the radius slider, the ionic radius of Metal 2 increases, and the displayed text reads Metal 1 has stronger metallic bonds than Metal 2. This explains how the increase in ionic radius affects the metallic bond.

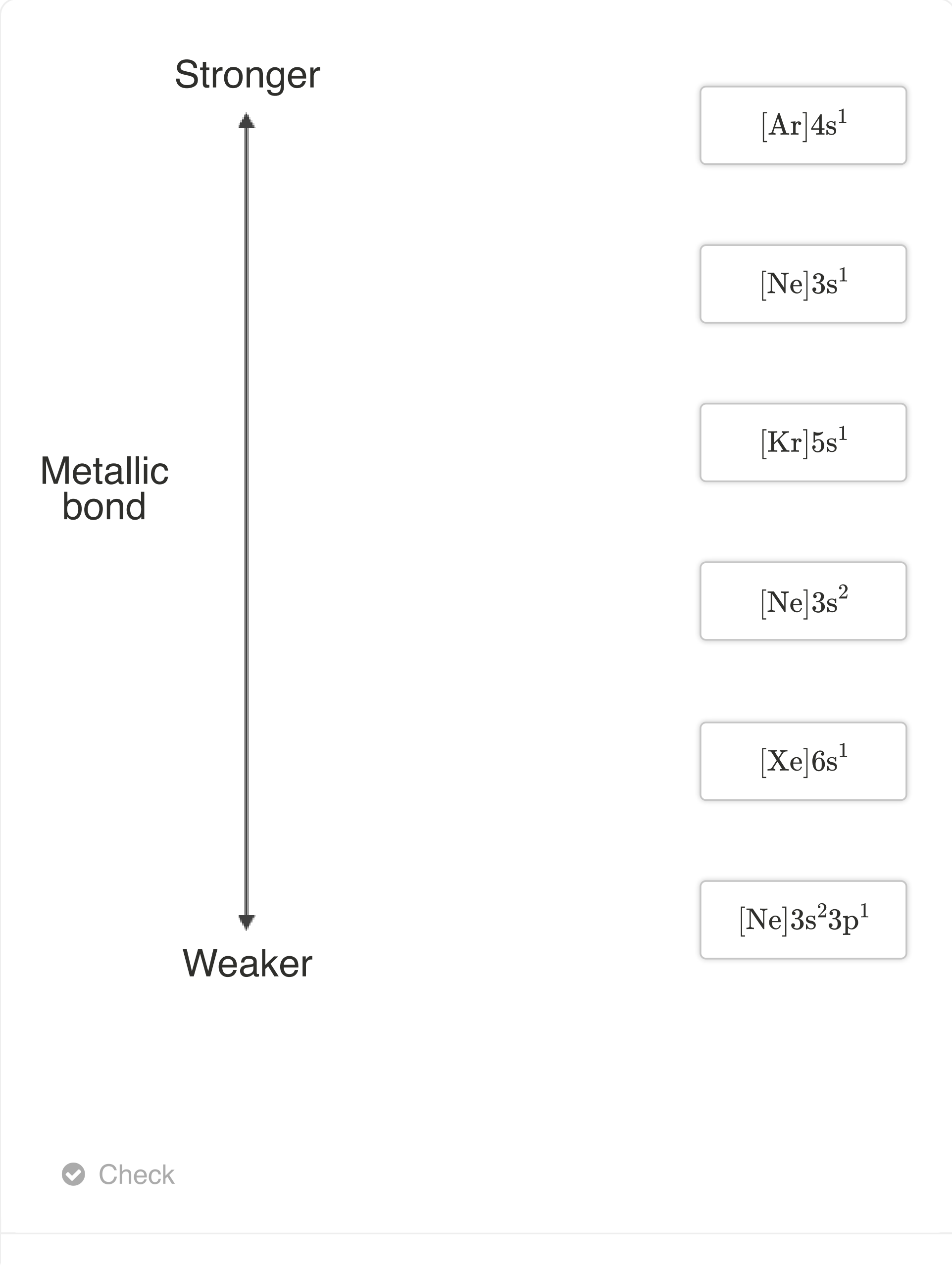
On moving the charge slider, the ionic charge of Metal 2 increases with the increase in the number of delocalized electrons, and the displayed text reads, Metal 1 has weaker metallic bonds than Metal 2. This explains how the increase in charge affects the metallic bond.

This interaction helps users understand the influence of metal ion radius and charge on metallic bond strength.

Making connections

The strength of the metallic bond is dependent on the same factors as the strength of the ionic bond, as explored in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/). The strength of both bonds depends on ionic radius and ionic charge.

Check your understanding of the strength of metallic bonds with **Interactive 2** by dragging and dropping the boxes of different electron configurations of metals and placing them in order of the strength of metallic bond from strongest to weakest.



Interactive 2. Understanding the Strength of Metallic Bonds.

Strength of metallic bond and melting points

The strength of metallic bonds results in high melting points for metals. From your experience, what state of matter are metals at room temperature? Do you know of any metal that exists in a different state at room temperature? Why might that be?

At room temperature, all metals, except mercury, are solids. For metals found in the s and p block regions of the periodic table, the melting point is directly related to the strength of the metallic bond. Note that the metals in the d block are not considered in this section due to the added complexity when considering the valence electrons in d orbitals. Melting points for all elements can be found in section 8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/) of the DP chemistry data booklet.

Figure 4 shows a periodic table with melting points for the s and p block metals highlighted. Looking at groups 1 and 2, what conclusion could you draw about the melting point of metals as you go down the group? Why do you think this is? Looking at period 3, what conclusion could you draw about the melting point of metals as you travel across a period? Why do you think this is?

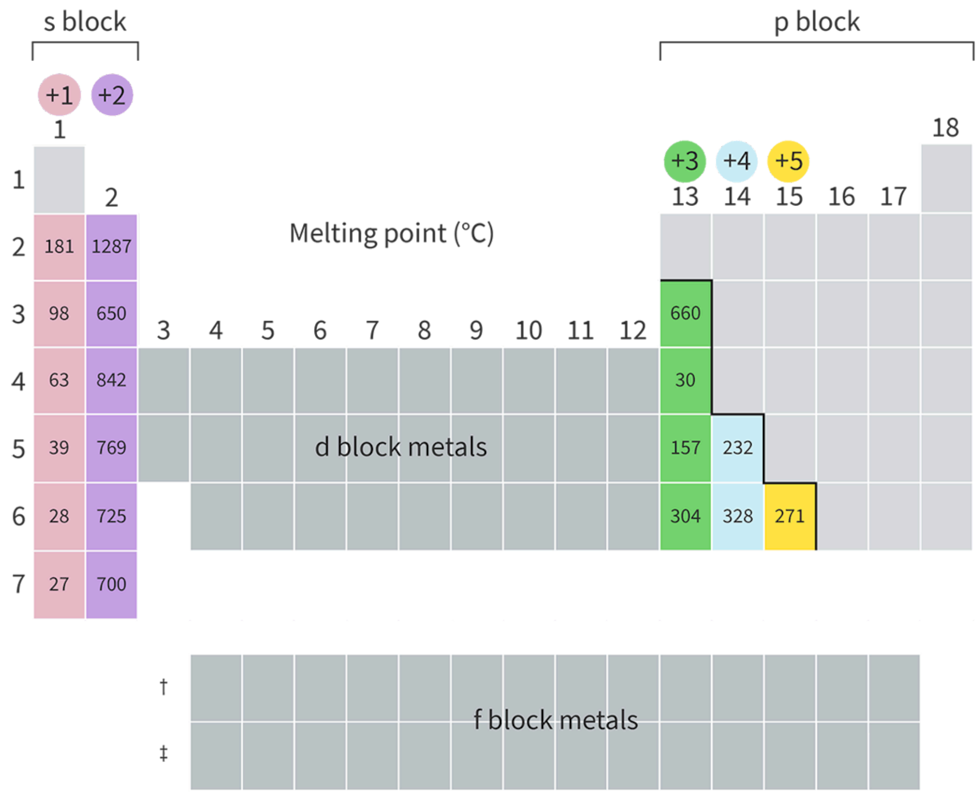


Figure 4. Melting points for the s and p block metals.

More information for figure 4

The image is a modified periodic table highlighting the melting points of s and p block metals. The table showcases groups 1 and 2, labeled with their respective changes in melting points as you move down the groups. For instance, Group 1 has melting points such as 181, 98, 63, 39, 28, and 27 moving downwards, indicating a general decrease. In Group 2, the melting points are 1287, 650, 842, 769, 725, and 700 moving downwards, showing variations. The s block metals are highlighted on the left side in shades of pink and purple. The p block metals on the right side include melting points such as 660, 30, 157, 304, 328, 232, and 271 in green, blue, and yellow blocks for typical groups +3, +4, and +5 respectively. The d and f blocks are shown in grey but not highlighted as they are not the primary focus. This visual representation indicates that melting points decrease downwards in groups and vary across periods due to differences in metallic bonding and electron density.

[Generated by AI]

As you can see, the melting point decreases as you go down a group of elemental metals. This is due to the decreased strength of the metallic bond as the radius of the metal ion increases. Generally, across a period, for elemental metals, the melting point increases. This is due to the increased strength of the metallic bond due to the increase in electron density as the charge on the metal ion becomes larger.

You might notice that the trend in melting point is much less distinct across a period than down a group. If you look at period 4, gallium is a significant outlier with a very low melting point: to the extent that objects made from gallium can melt when placed in the palm of your hand, or in warm water, as shown in **Video 1** below.

Video 1. Gallium spoon melting.

More information for video 1

The video presents a demonstration of a physical property of gallium metal. A transparent glass mug filled halfway with water is placed on a wooden table, alongside a polished, silver-coloured teaspoon made of gallium.
A hand enters the frame and lifts the spoon, initially holding it horizontally before rotating it to a vertical position. The hand then carefully lowers the spoon into the glass of water, immersing it. As the immersion progresses, the spoon is swirled around in the water. Over time, subtle changes become visible: small droplets begin to form along the spoon's surface. These droplets gradually increase in size before detaching and falling away, revealing that the spoon is undergoing a change. Rather than remaining solid, the metal appears to be melting. Eventually, the entire spoon melts completely, disappearing into the water.
The interactive video illustrates the low melting point of gallium as the metal transitions from a solid to a liquid when immersed in a cup with water. Gallium has a very low melting point, and when placed in water that is even slightly warm or on the palm of a hand, it begins to liquefy.
By showcasing the transformation of gallium in real time, the video highlights how physical properties such as melting point can vary significantly between different elements.

Theory of Knowledge

Mercury was consumed by many powerful historical figures, believing that it would grant immortality. Today, we have a greater understanding of this mysterious substance as a lethal neurotoxin, resulting in symptoms such as insomnia, emotional changes and disturbances in sensations. Does our current knowledge of this dangerous substance change our perspective of historical events?

Study skills

While there are trends for melting point in certain regions of the periodic table, there are not any trends for melting point across the periodic table. This is important to consider. The melting point trends discussed here are valid for s and p block metals, but do not apply to non-metals. It is important to consider the structure and bonding of elements to explain any trends.

Strength of metallic bond and hardness

The strength of metallic bonds results in a greater hardness of the metal. Hardness is a measure of the ability of a material to resist deformation. Think of the metals you have encountered at home: would you consider them hard or soft? What do you think that implies about the strength of their metallic bonds? Did you know that some metals are so soft they can be cut with a knife? **Interactive 3** shows two group 1 metals being sliced, sodium and potassium. Before watching them be cut, make a prediction. Which metal do you expect to be harder and have stronger metallic bonds? Which metal do you expect to be softer and have weaker metallic bonds? Then watch each metal being cut to see if you can identify the softer metal (weaker metallic bonds).

Interactive 3. Cutting Sodium and Potassium, a Measure of Hardness.

More information for interactive 3

The interactive contains two videos titled Sodium and Potassium. The video on the left for sodium is titled Sodium metal is soft and squishy. The video demonstrates the physical properties of sodium metal. The sodium metal chunks are stored in a glass container filled with oil. A chunk of sodium is removed from the oil and is pressed with fingers, revealing no signs of crumbling or breaking. The chunk is then cut using a knife. The video shows that sodium is soft enough to be easily cut with a knife, revealing its shiny interior. The cut piece is stored in the glass container with oil again. The video on the right for potassium is titled Potassium metal is like weird butter. The video demonstrates the physical properties of potassium metal. Potassium metal is taken out of a small tube and is then cut with a knife. The video shows that potassium is soft and can be easily cut like butter, revealing a shiny interior. The metal is pressed with fingers, revealing signs of crumbling and breaking. The cut piece of metal, when added to a container with water, reacts immediately emitting smoke and flame. The interactive demonstrates the metallic nature of sodium and potassium.

Activity

- **IB learner profile attribute**: Communicators
- **Approaches to learning**: Communication skills — Clearly communicating complex ideas in response to open-ended questions
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** Individual activity

Spreadsheet plotting

In this activity you will be creating individual plots to show how the radius of the metal ion affects the melting point for group 1, group 2 and group 13 metals.

- Predict what you expect the shape of each graph to look like for each group. Create a sketch of what you think each graph will look like.
- Create individual plots showing the effect of the radius of the metal ion on the melting point for group 1, group 2 and group 13 metals using information found in section 8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/) of the DP chemistry data booklet.
- For each plot, consider the following questions:
    - Did this plot look similar to your sketch? Why or why not?
    - Are there any data points that did not align with your expectations? Try researching to see whether you can come up with any explanations.
    - How do the melting points of s and p block metals provide evidence for the strength of the metallic bond?

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S2.3 The metallic model

Transition metals (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Identify d-block metals.
- Explain the high melting point and electrical conductivity of the transition elements.
- Compare d-block metals to s- and p-block metals.

So far in this subtopic about metals, we have largely ignored a large part of the periodic table containing metals that we often encounter in everyday life. When you think about examples of metals, what names come to mind? Some examples of metals that many are familiar with include iron as a metal used for car bodies, copper used in most electrical wiring and tungsten used to make extremely hard cutting and drilling tools. What region of the periodic table (**Figure 1**) are these elemental metals located?

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.01						Atomic number Element Relative atomic mass											2 He 4.00
2	3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
3	11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
4	19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
5	37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.96	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La † 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
7	87 Fr (223)	88 Ra (226)	89-103 Ac ‡ (227)	104 Rf (267)	105 Db (268)	106 Sg (269)	107 Bh (270)	108 Hs (269)	109 Mt (278)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (288)	116 Lv (293)	117 Ts (294)	120 Og (294)
			†	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97	
			‡	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)	

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Figure 1. Periodic table.

More information for figure 1

This image is a color-coded periodic table of elements. The table is divided into sections by colors, each representing different element groups. Elements are arranged in rows and columns. Each element's cell contains its atomic number, chemical symbol, and relative atomic mass. The color coding includes: alkali metals in orange, alkaline earth metals in yellow, transition metals in blue, lanthanides and actinides in red, metalloids in purple, nonmetals in pale green, halogens in green, and noble gases in violet. The table visually separates the lanthanides and actinides into a section below the main body of the table.

[Generated by AI]

Think back to when you first learned about electron configurations in section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/). What was it about the elements in groups 3–12 that were different to the elements in groups 1 and 2? Recall from section S2.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/) that transition metals are capable of forming ions of different charges. For example, two common ions of copper include Cu^{+} and Cu^{2+}. In section S3.1.8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/)you learned that the transition metal ions form complex ions through covalent bonding of a ligand. Because these metals have valence electrons occupying d orbitals, many of their properties are unique in comparison to their s and p metal counterparts. In this section, we will look more closely at the metals located in this d block region of the periodic table and their associated properties.

Making connections

In section S3.1.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/), periodic trends will be explored in more detail for s and p block elements. Why are periodic trends of the transition elements less evident? What is unique to the electronic components of these elements in comparison to the s and p block elements?

Recall from section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) that transition elements have valence electrons occupying d orbitals. Metals in this region of the periodic table are consequently referred to as d block metals. **Figure 2** shows an example of a few of

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these transition elements with their associated electron configuration. Note the terms ‘transition elements’ and ‘transition metals’ will be used interchangeably here as transition elements are metals.

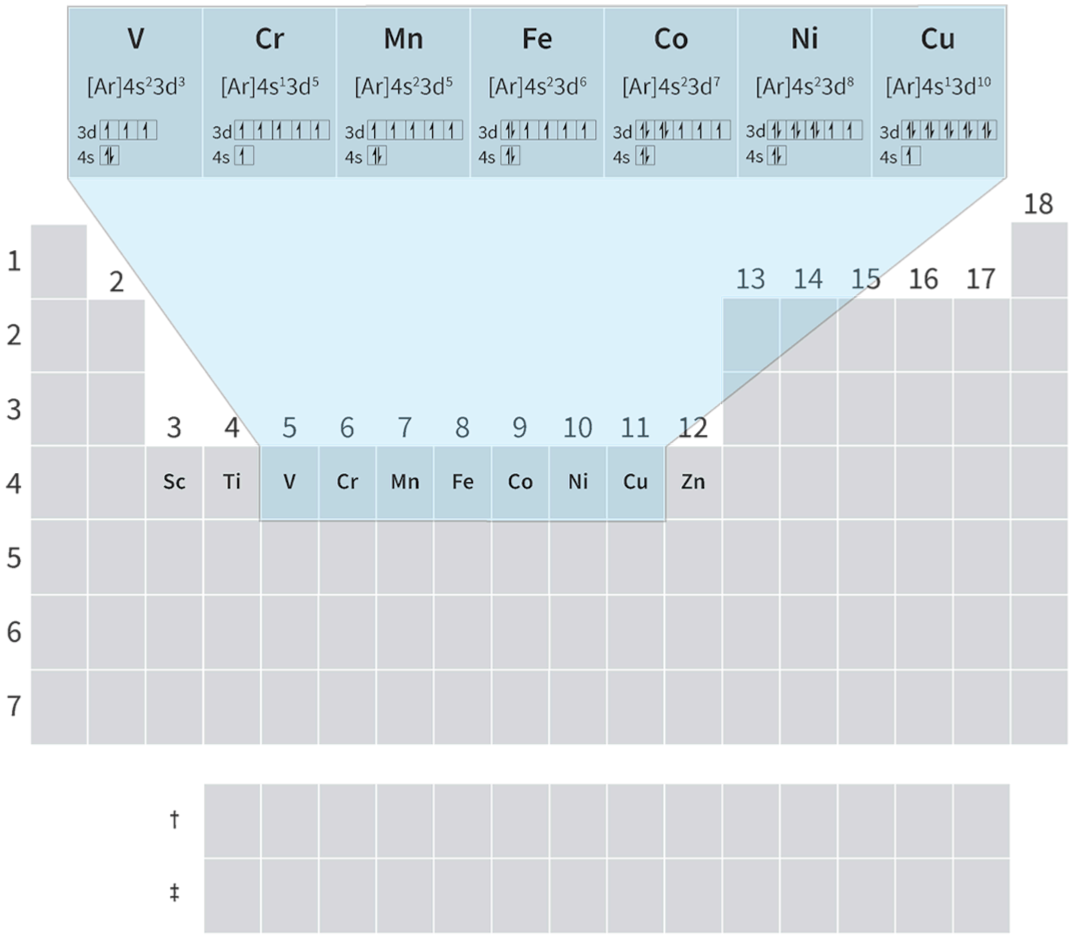

Figure 2. Some transition elements and their electron configurations.

More information for figure 2

This image is a section of the periodic table highlighting some transition elements along with their electron configurations. The section covers elements from group 5 to 12, which include V (Vanadium), Cr (Chromium), Mn (Manganese), Fe (Iron), Co (Cobalt), Ni (Nickel), and Cu (Copper). Above each element's symbol, their electron configuration is specified, showing the occupation of electrons in the d and s orbitals. For example, Vanadium (V) is shown to have the electron configuration [Ar] $4s^2$ $3d^3$. Each element in this section is visually represented along with a pattern illustrating their electron distribution. The image highlights the placement and structure of these transition metals within the periodic table, providing a visual understanding of their electron configuration specifics.

[Generated by AI]

For transition metals, the valence electrons can be considered to occupy both the s and d orbitals. For example, transition elements located in period 4 have valence electrons occupying 4s and 3d orbitals. This means that there are many possible

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ionic charges transition metals can form, as we saw in section S2.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/). These electrons are loosely bound and delocalise throughout the metallic lattice.

As there are a large number of valence electrons from both the s and d orbitals, this results in a greater electron density within the metallic lattice. This increased electron density in turn increases the strength of the metallic bond, as we saw in section S2.3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/), in comparison to s and p block metals.

Melting point

The melting points for period 4 metals are shown below in **Figure 3**. Notice how, in general, the melting points are much higher for the transition elements. This is due to the increased strength of the metallic bond. The large number of delocalised electrons from the 4s and 3d orbitals, increases the attraction they have with the metal ions in the lattice. Can you identify any metals in the d block that are not consistent with this trend of high melting point?

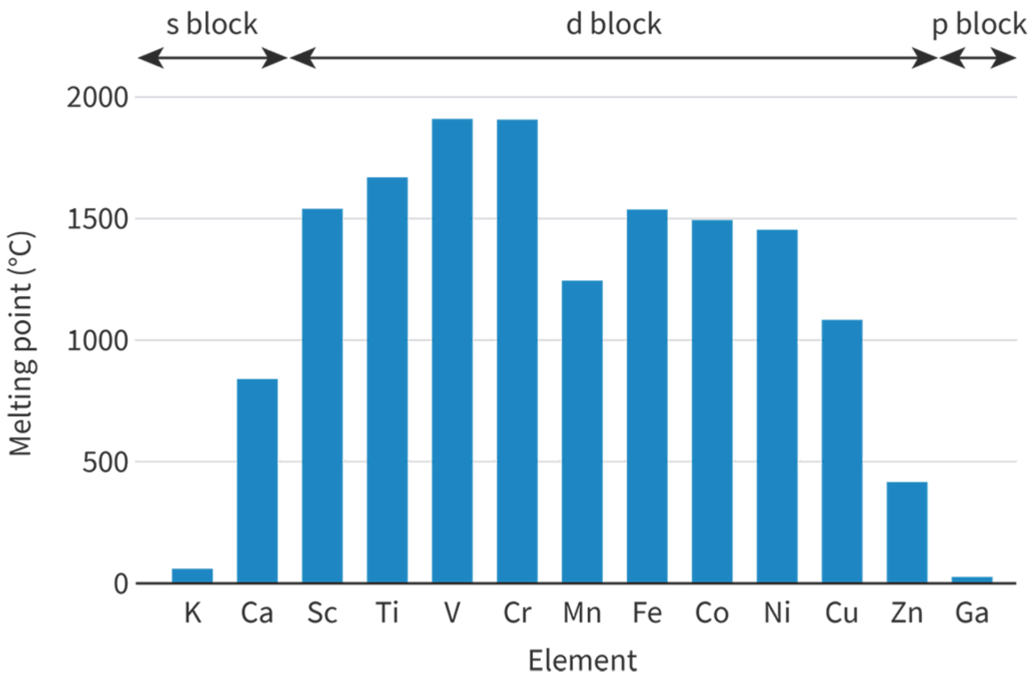

Figure 3. Melting points for period 4 metals.

More information for figure 3

The image is a bar graph showing the melting points of period 4 metals. The x-axis is labeled with the names of the metals: potassium (K), calcium (Ca), scandium (Sc), titanium (Ti), vanadium (V), chromium (Cr), manganese (Mn), iron (Fe), cobalt (Co), nickel (Ni), copper (Cu), and zinc (Zn). The y-axis represents the melting point in degrees Celsius, ranging from 0 to 2000 degrees Celsius at intervals of 500. Tallest bars are seen for chromium (Cr) and vanadium (V) which have high melting points above 1500 degrees

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Celsius. Notably, zinc (Zn) has a much lower melting point, depicted with a shorter bar. The graph highlights the trend where transition metals generally have higher melting points compared to other metals in period 4, except zinc.

[Generated by AI]

You might notice that zinc has a much lower melting point than the rest of the d block metals in period 4. In section S3.1.8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/) you will learn more about transition elements. There you will learn about why zinc, although it is a d block metal, is not technically considered a transition element.

Theory of Knowledge

You have just learned the relationship between the melting point of a metal and electron density. The melting point across the period 4 metals has many inconsistencies in this trend. How do inconsistencies in trends reduce the reliability of relationships?

Hardness

You saw earlier in section S2.3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/) how metals in group 1 were quite soft and could be easily sliced into. How would you expect transition elements to compare in hardness to metals in group 1? **Interactive 1** below shows someone cutting into potassium and copper, both period 4 metals. Are you able to identify the hard metal? Why do you think this is so?

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Interactive 1. The Varying Hardness of Cutting Potassium and Copper.

More information for interactive 1

Two videos titled Potassium and Copper. The video on the left, titled Potassium Metal is Like Weird Butter, focuses on potassium. The video demonstrates the physical properties of potassium metal. Potassium metal is taken out of a small tube and is then cut with a knife. The video shows that potassium is soft and can be easily cut like butter, revealing a shiny interior. The metal is pressed with fingers, revealing signs of crumbling and breaking. The cut piece of metal, when added to a container with water, reacts immediately emitting smoke and flame.

The video on the right for copper is titled Copper Sheet Thickness Guide 22 Mil (16 ounce copper). It provides a reference for the thickness of 22 mil copper sheeting, also known as 16 ounce copper. The video shows the copper sheet being folded easily. A scissor is used to cut the sheet and in to pieces. The video demonstrates the physical characteristics and flexibility of this specific copper thickness, offering users a better understanding of its properties. The video also directs users to a related thickness guide for more information.

Because copper has both 4s and 3d electrons as delocalised valence electrons, there is an increased attraction between these electrons and the metal ions in the lattice. This increased strength of metallic bonds results in a greater hardness of the copper in comparison to potassium.

Electrical conductivity

Transition elements are electrically conductive. These metals form their metallic bonds through the delocalisation of electrons in unfilled d orbitals. The electrostatic attraction between metal ions in the lattice and delocalised electrons increases with an increasing number of electrons in d orbitals. As there is free movement of electrons within these d orbitals, the transition metals are good conductors of electricity. Copper, for example, is one of the most conductive metals. For this reason, it is used in most electrical wiring as we saw in one of the examples earlier in **Figure 3**.

The properties discussed above are desirable properties for metals that allow it to be used in coins, jewellery, electrical wires, cookware, etc. **Interactive 2** shows common metal materials made from transition elements.

Rights of use

Interactive 2. Common Metal Objects Made of Transition Metals.

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair activity

1. Create a Venn diagram comparing and contrasting the properties of transition metals vs. s block metals. Consider use of the data booklet or researching data to help support your entries into the Venn diagram.
2. Design your own question to link this subtopic to another you have already learned about or to the nature of science.